

Experiment (12)

Network Design Using GNS3 Emulator

12.1 Objectives

To achieve usage in GNS3 Emulator

12.2 Equipment / Program Requirements:

GNS3 Emulator

12.3 Experiment Setup:

- 1- open gns3
- 2- draw topology
- 3- ping from device1 to device2

12.4 Simulation Results:

```

R1
R2
PC1
PC2
+

C 10.0.0.0 is directly connected, Ethernet0/0
  12.0.0.0/30 is subnetted, 1 subnets
C   12.0.0.0 is directly connected, Serial0/0
R1(config)#ip route 10.0.0.0 255.255.255.0 s0/0
R1(config)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

  10.0.0.0/24 is subnetted, 1 subnets
C   10.0.0.0 is directly connected, Ethernet0/0
  12.0.0.0/30 is subnetted, 1 subnets
C   12.0.0.0 is directly connected, Serial0/0
R1(config)#ip route 11.0.0.0 255.255.255.0 s0/0
R1(config)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

  10.0.0.0/24 is subnetted, 1 subnets
C   10.0.0.0 is directly connected, Ethernet0/0
  11.0.0.0/24 is subnetted, 1 subnets
S   11.0.0.0 is directly connected, Serial0/0
  12.0.0.0/30 is subnetted, 1 subnets
C   12.0.0.0 is directly connected, Serial0/0
R1(config)#
solarwinds | Solar-PuTTY free tool
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2 errors 3 warnings
10:07 AM
4/27/2026
DELL
```

```
relay ARG ...
rlogin [ip] port
save [FILENAME]
set ARG ...
show [ARG ...]
sleep [seconds] [TEXT]
trace HOST [OPTION ...]
version

Configure packet relay between UDP ports. See relay ?
Telnet to port on host at ip (relative to host PC)
Save the configuration to the file FILENAME
Set VPC name and other options. Try set ?
Print the information of VPCs (default). See show ?
Print TEXT and pause running script for seconds
Print the path packets take to network HOST
Shortcut for: show version

To get command syntax help, please enter '?' as an argument of the command.

PC1> trace 11.0.0.2
Trace to 11.0.0.2, 8 hops max, press Ctrl+C to stop
 1  10.0.0.1  15.201 ms  15.123 ms  15.417 ms
 2  *10.0.0.1  15.464 ms (ICMP type:3, code:1, Destination host unreachable)

PC1> ping 11.0.0.2
*10.0.0.1 icmp_seq=1 ttl=255 time=15.263 ms (ICMP type:3, code:1, Destination host unreachable)
*10.0.0.1 icmp_seq=2 ttl=255 time=15.208 ms (ICMP type:3, code:1, Destination host unreachable)
*10.0.0.1 icmp_seq=3 ttl=255 time=16.126 ms (ICMP type:3, code:1, Destination host unreachable)
*10.0.0.1 icmp_seq=4 ttl=255 time=15.987 ms (ICMP type:3, code:1, Destination host unreachable)
*10.0.0.1 icmp_seq=5 ttl=255 time=15.979 ms (ICMP type:3, code:1, Destination host unreachable)

PC1> ping 11.0.0.2
*10.0.0.1 icmp_seq=1 ttl=255 time=15.266 ms (ICMP type:3, code:1, Destination host unreachable)
*10.0.0.1 icmp_seq=2 ttl=255 time=15.736 ms (ICMP type:3, code:1, Destination host unreachable)
*10.0.0.1 icmp_seq=3 ttl=255 time=15.872 ms (ICMP type:3, code:1, Destination host unreachable)
*10.0.0.1 icmp_seq=4 ttl=255 time=15.751 ms (ICMP type:3, code:1, Destination host unreachable)
*10.0.0.1 icmp_seq=5 ttl=255 time=15.891 ms (ICMP type:3, code:1, Destination host unreachable)

PC1> ping 11.0.0.2
11.0.0.2 icmp_seq=1 timeout
84 bytes from 11.0.0.2 icmp_seq=2 ttl=62 time=76.938 ms
84 bytes from 11.0.0.2 icmp_seq=3 ttl=62 time=46.645 ms
84 bytes from 11.0.0.2 icmp_seq=4 ttl=62 time=61.220 ms
84 bytes from 11.0.0.2 icmp_seq=5 ttl=62 time=60.986 ms

PC1> 
```

R2

PC1

PC2

to Virtual PC Simulator, version 0.6.2
d to lling.

me: Apr 10 2019 02:42:20
t (c) 2007-2014, Paul Meng (mirnshi@gmail.com)
ts reserved.

free software, distributed under the terms of the "BSD" licence.
ode and license can be found at vpcs.sf.net.
information, please visit wiki.freecode.com.cn.

' to get help.

g the startup file

address 11.0.0.2 255.255.255.0 11.0.0.1

address

11.0.0.2 255.255.255.0 11.0.0.1

g for duplicate address...

11.0.0.2 255.255.255.0 gateway 11.0.0.1

ng 10.0.2

s from 10.0.0.2 icmp_seq=1 ttl=62 time=61.067 ms
s from 10.0.0.2 icmp_seq=2 ttl=62 time=47.179 ms
s from 10.0.0.2 icmp_seq=3 ttl=62 time=46.902 ms
s from 10.0.0.2 icmp_seq=4 ttl=62 time=92.064 ms
s from 10.0.0.2 icmp_seq=5 ttl=62 time=61.923 ms

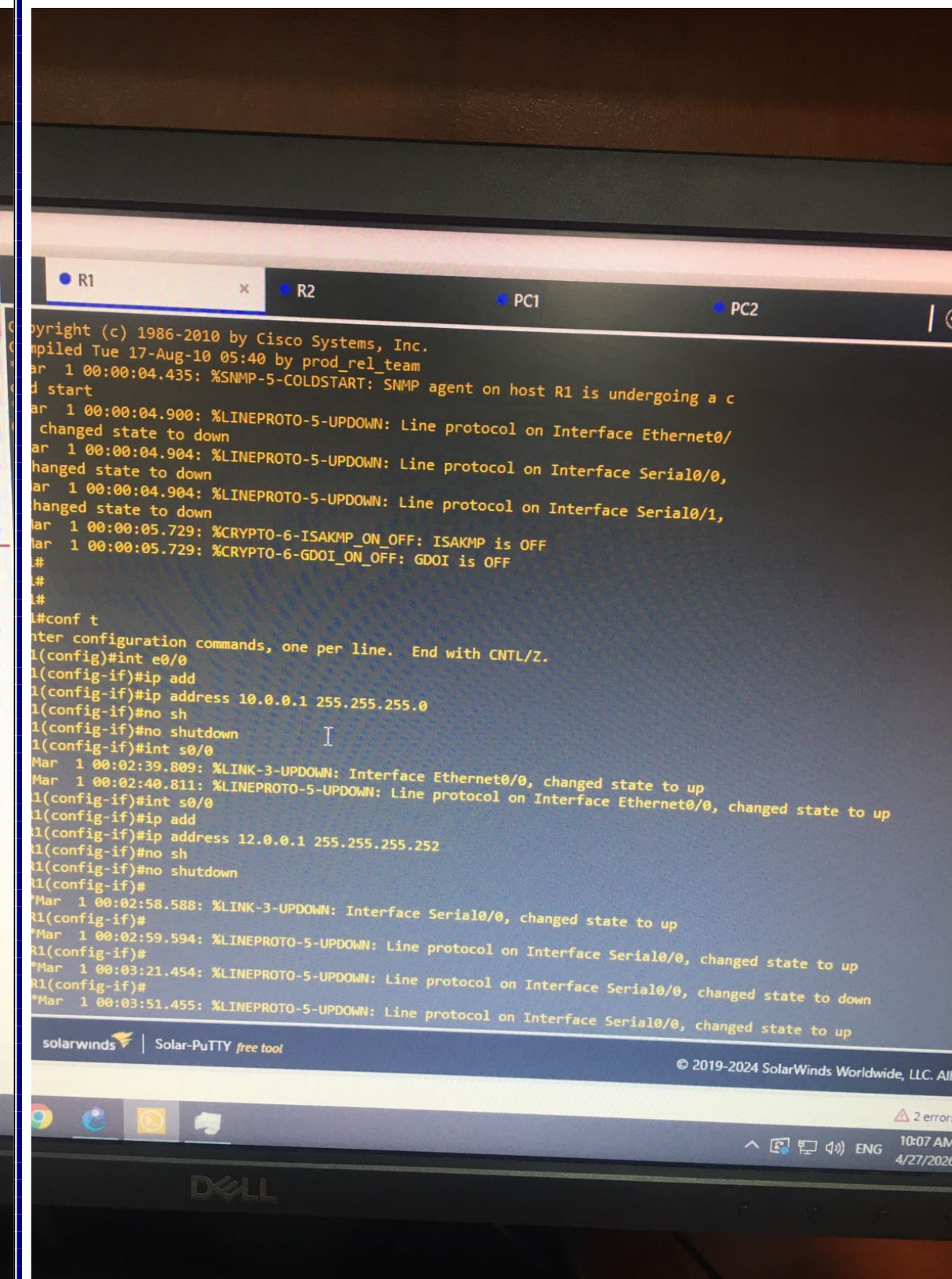
Solar-PuTTY free tool

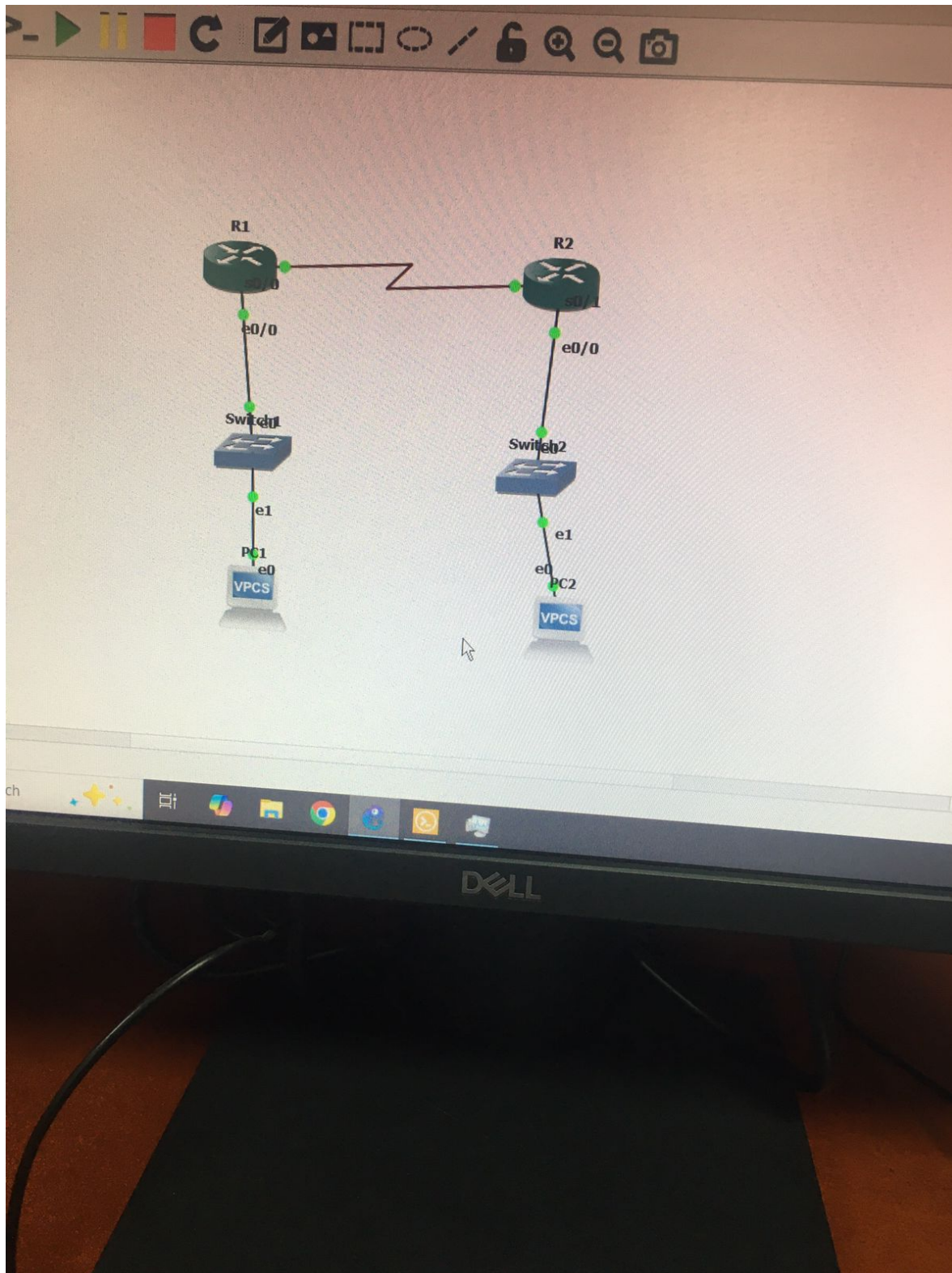
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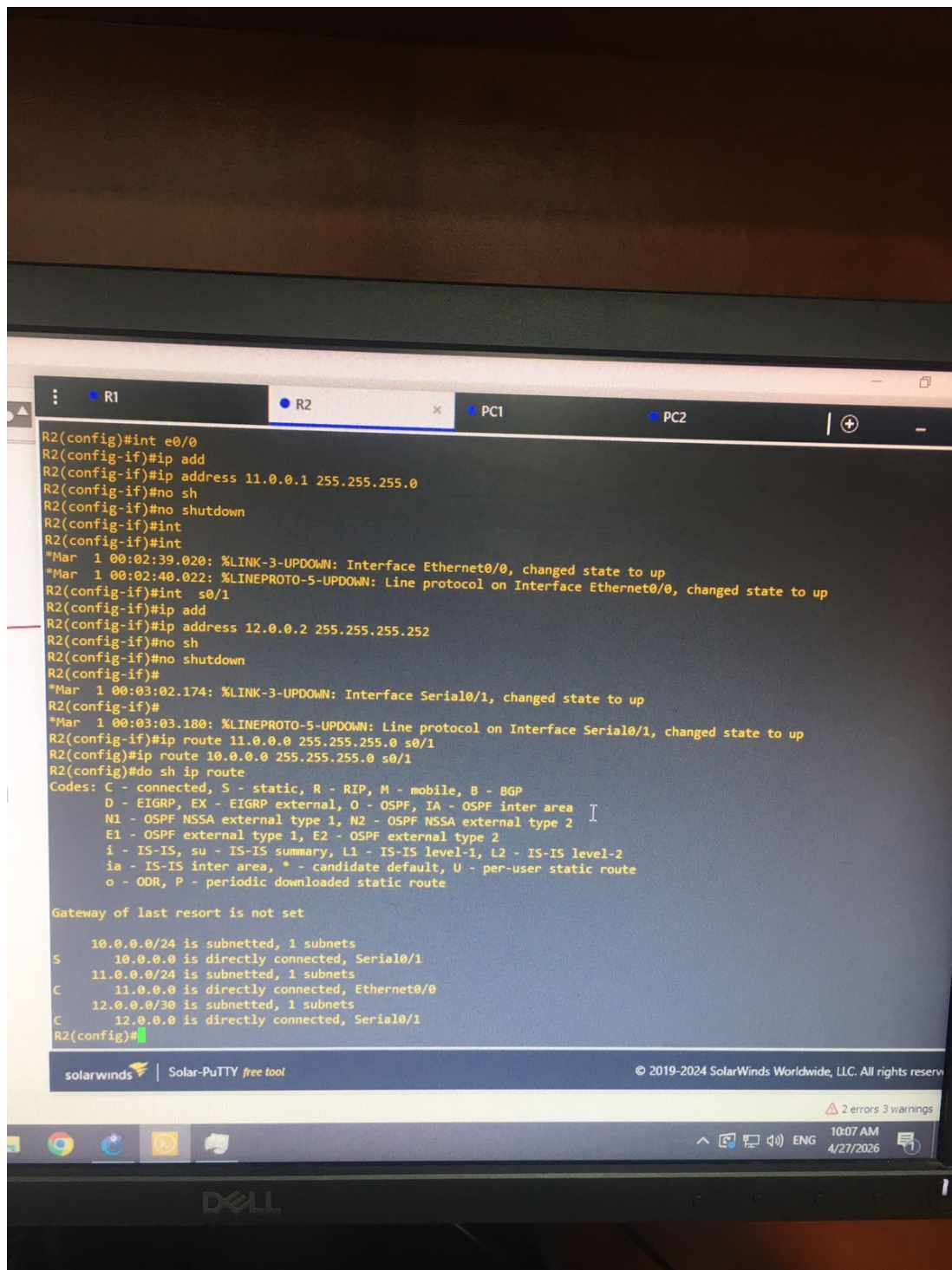
2 errors 3 warnings

ENG 10:07 AM
4/27/2026

DELL







12.5 Conclusions:

In conclusion, GNS3 is a powerful and flexible network simulation platform that allows users to design, configure, and test complex network topologies in a virtual environment. It provides a realistic experience by integrating real network operating systems, which makes it especially valuable for students and network engineers. Additionally, GNS3 supports a wide range of devices and protocols, enabling deep understanding of networking concepts without the need for expensive physical hardware